



Speech Transcription

SAKURA AI Engine Practical Exercise Module 2

<https://www.sakura.ad.jp/>

DATE

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COMPANY

SAKURA internet Inc.

SERVICE

SAKURA AI Engine

VERSION

Ver.1.0

Today's Agenda

SAKURA AI Engine Practical Exercise consists of four modules, 1 through 4.

Module 1	SAKURA AI Engine Practical Exercise Module 1 Log in, select a model, issue an Account Token, run inference, and build a RAG chatbot using a document.	
Module 2	SAKURA AI Engine Practical Exercise Module 2 Converting audio to text (transcription): Comparing accuracy in English and Thai to summarize	★ You are here
Module 3	SAKURA AI Engine Practical Exercise Module 3 AI output design that combines images and text to support business decision-making	
Module 4	SAKURA AI Engine Practical Exercise Module 4 Embed image and text AI calls into a web app and run it in a browser.	

Goals of this class

You can convert audio into text and verify for yourself whether it is done correctly.

1 Run the speech transcription API using curl

Passing the audio file to the API and converting it to text

2 Explain the process and limitations of passing audio to the API

Download the audio to your computer before sending it to the API

3 Check the difference in accuracy between English and Thai

This is the most important goal in this class

4 Explain how errors affect the summary

Compare the summary before and after fixing the errors

Proceed in the order of sample audio → language accuracy comparison → summary.

Timetable (90 minutes)

After moving with sample audio, compare the accuracy of English and Thai, and finally connect it to a summary.

Time	Contents
0 - 10	Overview (1) What is Voice AI?
10 - 25	Hands-On 1 Transcribe sample audio
25 - 33	Overview (2) Process of passing voice to API
33 - 55	Hands-On 2 Compare accuracy in English and Thai
55 - 63	Overview (3) Transcription accuracy
63 - 85	Hands-On 3 Check the transcription and summarize
85 - 90	Summary & Upcoming Information

How to proceed with this class

During the process, you will see supporting websites and recording apps. For the slides, show "What will you do now?"

What you use

- Slides ... Overview and what to do now
- Website ... Procedures, commands, worksheets
- Recording app ... Record in your browser and generate a URL

Required and Optional

- Each hands-on session consists of two stages: mandatory and advanced
- Once the required steps are complete, proceed directly to Optional
- You can proceed at your own pace

If You Get Stuck

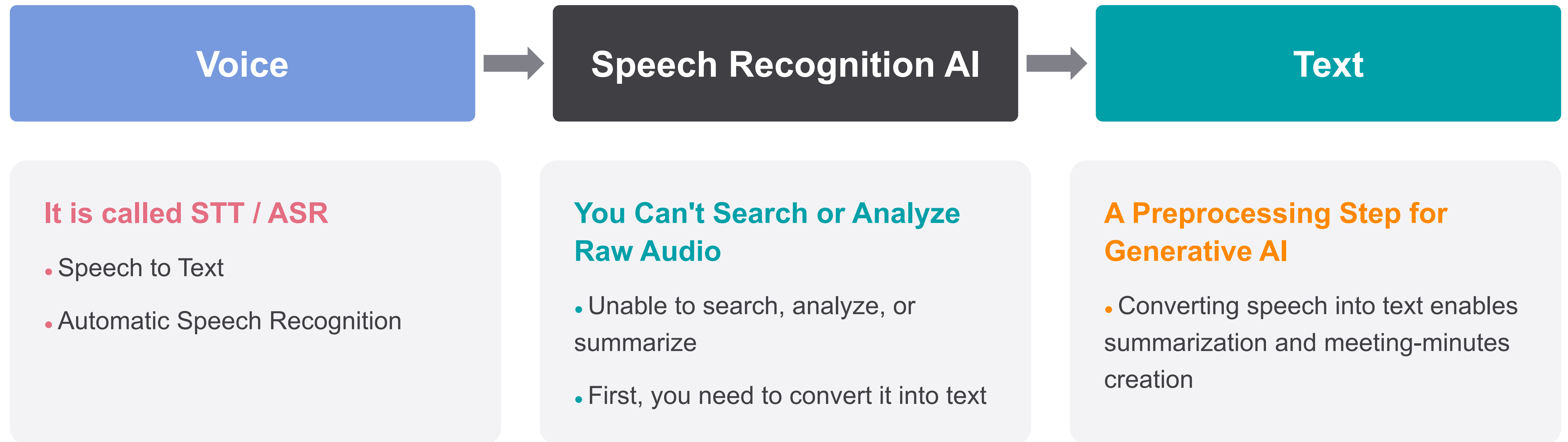
- Find error messages with Troubleshooting
- You may consult with someone nearby
- If the issue is not resolved, ask the instructor or a TA

Overview (1) (0-10 minutes)

What is Voice AI?

What is Voice AI (Transcription)?

This technology analyzes spoken speech and outputs the content as text.



In this class, we use a speech recognition model called Whisper.

Model used this time: Whisper

SAKURA AI Engine uses a speech recognition model called Whisper.

whisper-large-v3-turbo

- OpenAI released a speech recognition model
- A lighter derivative model of the large-v3

Structure focused on speed

- Faster inference than the original model
- Open source and publicly available

How are long audio handled?

Whisper internally processes audio in approximately 30-second increments.

SAKURA AI Engine automatically splits and processes long audio, so users can send audio up to 30 minutes or 30MB.

* 30 seconds is Whisper's internal processing unit and is not the upload limit.

What is diarization?

This is a technique for identifying who spoke. It is different from transcription.

For transcription only,

- You understand what was said
- You don't know who made each statement
- Everyone's statements become a single text

When there is speaker diarization,

- Separate speakers for each statement
- Can be organized in the format A / B
- You can retrieve the "person responsible" from the minutes

Remember:

The Whisper we use this time does not support speaker diarization.

When multiple people record together, everyone's statements are returned as a single transcript.

Real-time processing and batch processing

There are two ways to convert audio to text when to do so.

Real-time processing

- Write down words on the spot while speaking
- Use for subtitles, simultaneous interpretation, and more
- The system tends to become complex

Batch processing ★ This time is here

- After recording, put it all together into text
- Suitable for preparing meeting minutes
- Simple mechanism

The API used this time is a synchronous batch processing method. When you send a file, the result is returned after the processing is complete.

Factors that affect recognition accuracy

Even with the same AI, the results can vary greatly depending on the conditions.

Sound Quality

- Surrounding noise
- Distance from the microphone

Way of Speaking

- Speaking Speed
- Volume and overlap of voice

Words

- Technical Terms & Proper Nouns
- Abbreviations and Speech Delays

Language

- Differences between English and Thai
- Mixing multiple languages

Points for Business Use

Preparing meeting minutes / Call center call records / Video subtitle creation

All of these address the problem of "not being able to search or analyze using voice alone."

MISSION 01

Transcribe sample audio

Required

1. Create a working folder 02_AI_ASR
2. Download a short sample audio (about 10 seconds)
3. Run the Transcription API with curl
4. Run long sample audio (about 1 minute) and compare results

Optional | Extra Challenge

- Compare results with noisy, fast-speaking samples

Start with audio that's known to work reliably. You'll do your own recording in the next hands-on session.

Before you start: Confirm your preparation

Hands-On 1 Transcribing sample audio

We will use the environment created in Module 1 as is.

Account Token

- Issued at Module 1

WSL (Ubuntu)

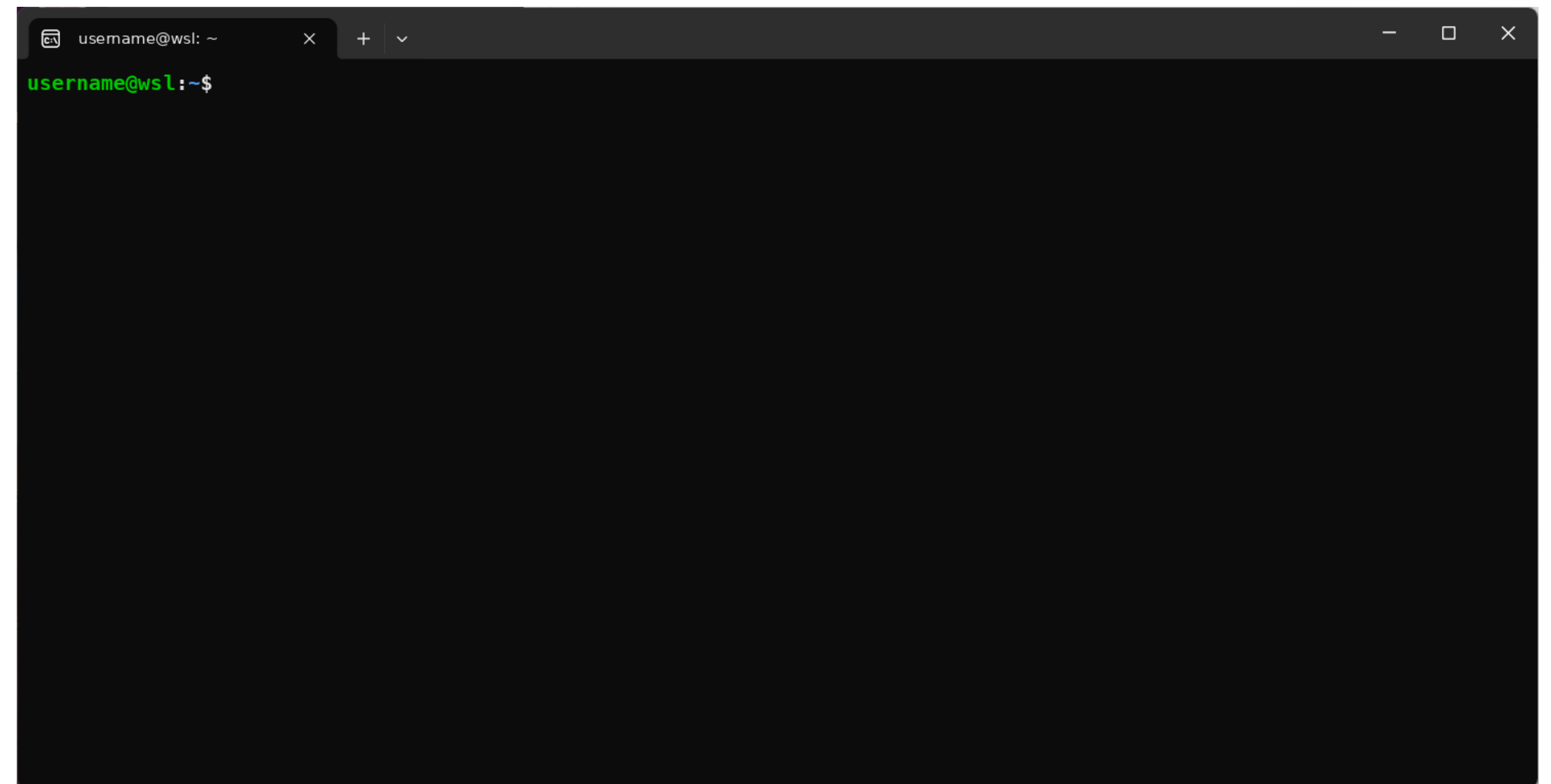
- Open the terminal

Check if tokens remain

```
echo $AI_ENGINE_TOKEN  
# If nothing appears, run export again
```

If it disappears, set it again

```
export AI_ENGINE_TOKEN="Account Token issued <>"
```



About the sample audio used

Hands-On 1 Transcribing sample audio

First, use a sample that can be transcribed reliably. We have prepared the following three options.

sample_short_en.wav

- Short English audio (about 10 seconds)
- First, check how it works

sample_short_th.wav

- Short Thai audio (about 10 seconds)
- Hands-On 2 Comparison with English

sample_long_en.wav

- Long English audio (about 1 minute)
- Compare how the results will change

The audio files are available on GitHub.

You can download it directly with curl.

`https://vtmacs.github.io/ameicc2026/assets/downloads/< File name>`

Starting with your own recording introduces two possible sources of error, so start with audio that's known to work reliably first.

* Audio data source: Google FLEURS Dataset (CC BY 4.0), downloaded from <https://huggingface.co/datasets/google/fleurs>

STEP 1 | Prepare the work folder and sample audio

Hands-On 1 Transcribing sample audio

1 Create a work folder for this occasion
02_AI_ASR will be named

2 Download a short sample audio
sample_short_en.wav

3 Check if the file is created in ls

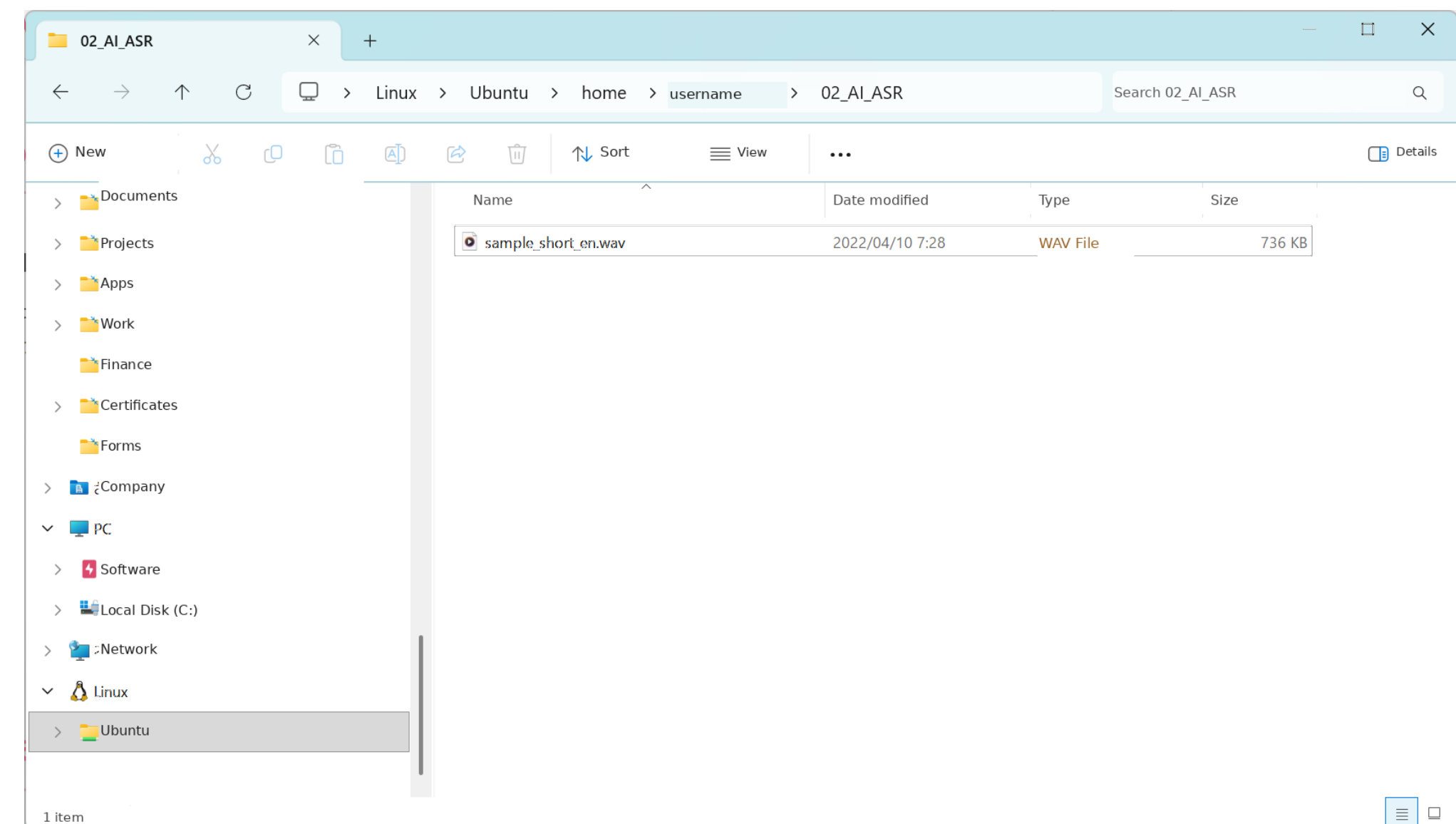
Key Points

Since the URL contains the file name,

You can save the name as is with -O (uppercase).

Running on WSL (Ubuntu)

```
mkdir -p ~/02_AI_ASR
cd ~/02_AI_ASR
curl -O ¥
    https://vtmacs.github.io/ameicc2026/assets/downloads/sample_short_en.wav
ls -l
```



Note: Distinguishing Between -O and -o

Hands-On 1 Transcribing sample audio

There are two similar options. You can use them differently in the form of URLs.

-O (capital O): Use this when the URL already contains a file name with extension.

```
curl -O https://vtmacs.github.io/ameicc2026/assets/downloads/sample_short_en.wav  
# → Saves the file using the name in the URL.
```

-o (lowercase o): Use this when you want to choose the file name yourself.

```
curl -o my_voice_en.mp3 https://ai-handson.f5.si/voice-uploader/s/xxxxxx  
# → Saves the file using the name you specify.
```

If the downloaded file does not have a proper file name or extension, download it again using -o.

STEP 2 | Run the Speech Transcription API

Hands-On 1 Transcribing sample audio

Send audio files and convert them to text

```
curl --request POST ¥  
  --url https://api.ai.sakura.ad.jp/v1/audio/transcriptions ¥  
  --header 'Accept: application/json' ¥  
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥  
  --header 'Content-Type: multipart/form-data' ¥  
  --form 'file=@sample_short_en.wav' ¥  
  --form 'model=whisper-large-v3-turbo' ¥  
  --form 'language=en'
```

1

endpoint

URL for speech transcription

2

Authentication

Pass the token of the environment variable

3

File

Sending the audio file itself

4

Model

whisper-large-v3-turbo

5

Language

Specify en

If you don't specify the language, it may not be able to transcribe properly. If it's English, always include 'en'.

Meaning of Request Parameters

Hands-On 1 Transcribing sample audio

item	Meaning
<code>--request POST</code>	Specifying the format in which data is sent
<code>--url</code>	Endpoints of the Speech Transcription API
<code>Authorization</code>	Authentication with Account Tokens
<code>Content-Type</code>	multipart/form-data (file submission format)
<code>file</code>	Transcribe audio file ← followed by the file name @
<code>model</code>	Model used for speech recognition (whisper-large-v3-turbo)
<code>language</code>	Speech language (en=English, th=Thai). If not specified, accuracy may decrease

Supported audio formats include mp3 and wav. Depending on the format and quality, it may not be transcribed correctly.

How to read Response (JSON)

Hands-On 1 Transcribing sample audio

```
{
  "text": "(The audio content was transcribed
            Text)",
  "model": "whisper-large-v3-turbo"
}
```

1

text

Here you will see the transcription results

2

model

Speech recognition models used

Success

Unlike the Chat Completions API used in Module 1, the result is returned directly in the "text" field.

STEP 3 | Review the results of short audio

Hands-On 1 Transcribing sample audio

Since the audio is about 10 seconds long, you can immediately check if it was taken correctly.

Is it just as I heard it?

- Does the content match what was read aloud?

Few punctuation marks and line breaks

- Spoken words come out exactly as they are

Mixed with misperception

- Pay special attention to technical terms and proper nouns

Original text of sample_short_en.wav (correct answer)

Japan holds nearly 7,000 islands (the biggest being Honshu), making Japan the 7th largest island in the world!

Answer checking

Compare this sentence with the text you received.

Are the numbers and proper nouns (7,000 / Honshu) correct?

STEP 4 | Run and compare even with long audio

Hands-On 1 Transcribing sample audio

Using the same command, run a sample audio of about one minute.

Run It with a Different File Name

```
curl -O https://vtmacs.github.io/ameicc2026/assets/downloads/sample_long_en.wav
curl ... --form 'file=@sample_long_en.wav' ¥
      --form 'model=whisper-large-v3-turbo' --form 'language=en'
```

Original text of sample_long_en.wav (correct answer)

Safaris are perhaps the greatest tourism draw in Africa and the highlight for many visitors. The Kruger National Park (KNP) lies in the north-east of South Africa and runs along the border of Mozambique in the east, Zimbabwe in the north, and the southern border is the Crocodile River. The park covers 19,500 km² and is divided in 14 different ecozones, each supporting different wildlife. Examples of on-site activities include hunting, fishing, photography, bird watching, and visiting parks and studying information about the ecosystem. Some animals, such as elephants and giraffes, tend to approach closely to cars and standard equipment will allow good viewing.

What you should see

- Are there any gaps along the way?

Differences from short audio

- Did the number of transcription errors increase?

The longer the audio, the harder it is to read as is. Here's why you can summarize it later.

Optional | Compare accuracy by changing conditions

Hands-On 1 Transcribing sample audio

Once you've finished the required tasks, try checking what changes in recognition accuracy.

1 Download two types of samples

2 Execute transcription with the same command
`--form 'language=en'`

3 Find out where it broke down

Key Points to Watch

- Words that are easily misunderstood
- Sentence Breaks

Download the Noisy / Fast-Speaking Samples

```
curl -O ¥  
  https://vtmacs.github.io/ameicc2026/assets/downloads/sample_long_en_noisy.wav  
curl -O ¥  
  https://vtmacs.github.io/ameicc2026/assets/downloads/sample_long_en_fast.wav
```

Common Errors and How to Fix Them

Hands-On 1 Transcribing sample audio

An error is not a failure. If you read the message, it usually indicates the cause.

No such file or directory

File not found

- Check the file name with LS
- Check if you have CD ~/02_AI_ASR
- Is the name after @ correct?

File name xxxxxx

Saved without extension

- Rename the file with -o

401 Unauthorized

Tokens are not being transferred correctly

- Confirm with echo
\$AI_ENGINE_TOKEN
- Check for unnecessary gaps before or after

Please look for error messages directly on the supporting website's Troubleshooting section.

Progress Check

Checkpoint

Before moving on, please check if you have made this far.



Created the work folder 02_AI_ASR



Could transcribe short sample audio



It was possible to execute even long sample audio



We were able to find a difference between the two outcomes.

How to proceed

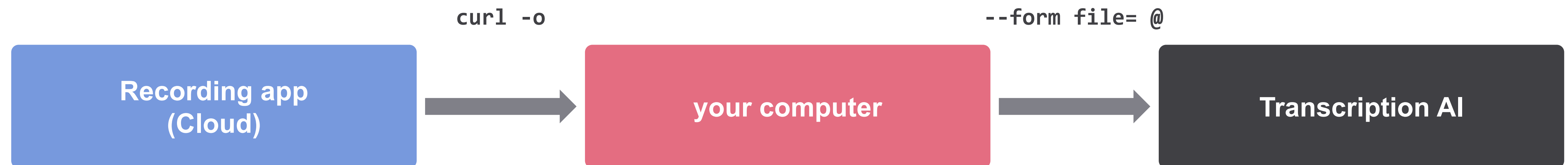
If you're already finished, go to Optional. If something isn't going well, please ask the instructor or their representative.

Overview (2) (25-33 min)

Process of passing voice to API

Process of passing voice to API

This API cannot accept a URL. It only accepts the file itself.



(1) Download

- Bring voice from the cloud to your fingertips

(2) Upload

- Send your files to the API

"To hand data to AI, you first need to bring that data to your own machine." This same constraint often comes up in real system design.

Synchronous and Asynchronous Processing

When asking AI to handle the process, there are two ways to wait.

Synchronous ★ Used in This API

- Send and wait until processing is complete
- Results come back exactly as they are
- It's simple, but not suitable for lengthy processing

Asynchronous Processing

- Once you send it, it's over for now
- Come back to get results later
- Suitable for long-term or large-scale processing

Synchronous processing has a limit on the "waiting time." That's why the next slide has a limit.

Length limit: 30 minutes or 30MB

Because this API is synchronous, there is a limit to the length of the voice that can be sent at once.

Maximum size per request

- Up to 30 minutes of audio
- Up to 30 MB

What is happening inside?

- Automatically splits and recognizes every 30 seconds, connecting and returning results.

Handling Longer Audio

A two-hour meeting cannot be sent in a single request.

Split the audio and send multiple requests

Use a separate asynchronous workflow

MISSION 02

Compare accuracy in English and Thai

Required

1. Transcribe English sample audio
2. Transcribing Thai sample audio
3. Count the number of misunderstandings by comparing with the script
4. Record the differences between English and Thai on a worksheet

Optional | Extra Challenge

- Record your own voice and try the same thing
- Record with two or three people to see whether you can tell who said what

Use the provided audio files. Recording your own voice is optional, if you have time.

About the voice used

Hands-On 2: Comparing accuracy in English and Thai

Two audio sets of the same content read aloud in English and Thai will be distributed.

sample_short_en.wav

- Audio reading the script content aloud in English

sample_short_th.wav

- Audio of the same content read aloud in Thai

Why Do the Samples Use the Same Content?

Since the content is the same, you can only compare differences in recognition accuracy depending on the language.

Matching the speaker, recording environment, and content is the basic basis of comparison.

Where to store audio files

<https://vtmacs.github.io/ameicc2026/assets/downloads/< File name>>

Original audio text (correct text)

Hands-On 2: Comparing accuracy in English and Thai

The English and Thai recordings contain the same content.
Use the following text as the reference.

sample_short_en.wav (English)

Japan holds nearly 7,000 islands (the biggest being Honshu), making Japan the 7th largest island in the world!

sample_short_th.wav (Thai)

ญี่ปุ่นมีเกาะเกือบ 7,000 เกาะ (ฮอนชูคือเกาะที่ใหญ่ที่สุด) ส่งผลให้ญี่ปุ่นเป็นเกาะที่ใหญ่ที่สุดอันดับ 7 ของโลก!

Verification Points

- Numbers (7,000 / 7)
- Proper Nouns (Honshu / ฮอนชู)

STEP 1 | Transcribing English audio

Hands-On 2: Comparing accuracy in English and Thai

(1) Download the English audio

```
cd ~/02_AI_ASR  
curl -O https://vtmacs.github.io/ameicc2026/assets/downloads/sample_short_en.wav
```

(2) Pass it to the transcription API

```
curl --request POST ¥  
  --url https://api.ai.sakura.ad.jp/v1/audio/transcriptions ¥  
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥  
  --header 'Content-Type: multipart/form-data' ¥  
  --form 'file=@sample_short_en.wav' ¥  
  --form 'model=whisper-large-v3-turbo' ¥  
  --form 'language=en'
```

STEP 2 | Transcribing Thai audio

Hands-On 2: Comparing accuracy in English and Thai

Repeat the same steps again with Thai audio.

Run It with a Different File Name

```
curl -O https://vtmacs.github.io/ameicc2026/assets/downloads/sample_short_th.wav  
curl ... --form 'file=@sample_short_th.wav' ¥  
        --form 'model=whisper-large-v3-turbo' --form 'language=th'
```

What you should see

- How correctly Thai was recognized
- How does it compare to the English result?

Because there is a script,

- Compare with the correct text for an objective comparison

STEP 3 | Count against the script

Hands-On 2: Comparing accuracy in English and Thai

Check if you got it correctly and count it by eye.

(1) Place the script and transcription results side by side

(2) Mark words that are mistakenly recognized

(3) Count the transcription errors

- Counted separately in English and Thai

Easier to Count and Compare

Writing it as "English: ○ errors / Thai: ○ errors" makes the difference clear.

```
username@wsl: ~/02_
username@wsl :~$ cd 02_AI_ASR/
username@wsl :~/02_AI_ASR$ curl --request POST --url https://api.ai.sakura.ad.jp/v1/audio/transcriptions --header
'Accept: application/json' --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" --header 'Content-Type: multipart/form-d
ata' --form 'file=@sample_short_en.wav' --form 'model=whisper-large-v3-turbo' --form 'language=en'
{"text": "Japan holds nearly 7,000 islands, the biggest being Honshu, making Japan the seventh largest island in the wor
ld.", "model": "whisper-large-v3-turbo"}
username@wsl :~/02_AI_ASR$
username@wsl :~/02_AI_ASR$ curl --request POST --url https://api.ai.sakura.ad.jp/v1/audio/transcriptions --header
'Accept: application/json' --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" --header 'Content-Type: multipart/form-d
ata' --form 'file=@sample_short_th.wav' --form 'model=whisper-large-v3-turbo' --form 'language=th'
{"text": "ญี่ปุ่นมีเกาะ เกือบ 7,000 เกาะ หอนชู คือเกาะที่ใหญ่ที่สุด สงพลให้ญี่ปุ่น เป็นเกาะที่ใหญ่ที่สุด ดันนับเจ็ดของโลก", "model": "whisper-large-v3-turbo"}
```


STEP 4 | Record on Worksheet

Hands-On 2: Comparing accuracy in English and Thai

Transcription (English)

English transcription results

Transcription (Thai)

Thai transcription results

Errors in English

Words and numbers misunderstood in English

Errors in Thai

Misidentified Words and Numbers in Thai

Think about it

Which language was correct? And why do you think that is?

You'll use this record again in the next hands-on session (Hands-On 3).

Why does accuracy vary depending on the language?

In general, the ease of recognition varies by language. The main reasons are as follows:

Volume of training data

- Overwhelmingly large amounts of data in English
- More training data tends to improve accuracy.

Word Boundaries

- Thai does not insert spaces between words
- It becomes harder to determine boundaries

Proper Nouns and Place Names

- Local names are especially easy to mishear
- Because they rarely appear in the training data

Sound characteristics

- Differences in tone and vowels
- Difficult to distinguish similar sounds

What matters in practice

When people talk about "high AI accuracy," they often refer to the language the model excels at.

Be sure to check the accuracy using the language you actually plan to use.

Optional | Preparing the recording

Hands-On 2: Comparing accuracy in English and Thai

If you have the energy, try using your own voice. You can record using your smartphone's browser.

1 Open the recording app page on your smartphone
<https://ai-handson.f5.si/voice-uploader/>

2 Enter your Basic authentication ID / password
This is listed on the supporting website.

3 Allowing the use of microphones
If you mistakenly refuse, you won't be able to record.

If you refuse the microphone,

Reset the site's microphone permissions and reopen the page.

The procedure is in Troubleshooting.

Language English ▾

🎤 Voice Uploader

Record with microphone → preview → upload
→ get a Short URL.

Start Recording

Ready

Optional | Record your own voice

Hands-On 2: Comparing accuracy in English and Thai

- 1 Receive the English script
- 2 Start recording and read aloud in 1–2 minutes
- 3 Stop recording and upload

Tips for Recording

- Hold your smartphone and bring it close to your mouth to speak
- Move to a quiet place
- Keep it to 1–2 minutes

Language English ▾

Voice Uploader

Record with microphone → preview → upload
→ get a Short URL.

Stop Recording

Recording...

Optional | Enter URL on PC

Hands-On 2: Comparing accuracy in English and Thai

1

Check the URL displayed after uploading

2

Write down the last six characters (letters and numbers) at the end

Example: .../voice-uploader/s/xxxxxx

3

Manually entering into the PC terminal

No transfer tools required

Note

Recorded data is deleted within 24 hours.

Please record the text on the worksheet, not the URL.

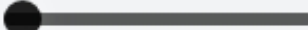
Language English ▾

Voice Uploader

Record with microphone → preview → upload
→ get a Short URL.

Start Recording

Upload complete!

▶ 0:00 / 1:13   ⋮

Upload this audio

Short Code: xxxxxx

Short URL

<https://ai-handson.f5.si/voice-uploa>

Copy

Optional | Download and transcribe

Hands-On 2: Comparing accuracy in English and Thai

(1) Download the recording to your computer.

```
cd ~/02_AI_ASR
curl -o my_voice_en.mp3 https://ai-handson.f5.si/voice-uploader/s/xxxxxx
ls -l
```

(2) Pass it to the transcription API

```
curl --request POST ¥
  --url https://api.ai.sakura.ad.jp/v1/audio/transcriptions ¥
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥
  --header 'Content-Type: multipart/form-data' ¥
  --form 'file=@my_voice_en.mp3' ¥
  --form 'model=whisper-large-v3-turbo' ¥
  --form 'language=en'
```


Optional | Try recording with 2 to 3 people

Hands-On 2: Comparing accuracy in English and Thai

If you succeeded on your own, try reading aloud together with a few people.

- 1 Divide roles into 2 to 3 people and read the script
- 2 Transcribe using the same steps
- 3 Check the results to see whether you can tell whose statement is whose

What you should notice here

All of our statements are returned as a single text.

There is no information on "who said it."

Optional | Verify by changing conditions

Hands-On 2: Comparing accuracy in English and Thai

If you have more time, look into what makes your accuracy change.

mix languages

- Try speaking a mix of English and Thai

Include a proper noun

- See How University or Product Names Get Misrecognized

Change location

- Compare recordings in a quiet place and inside the classroom

Change the distance

- Compare bringing the microphone close versus keeping it at a distance

If the results are poor, try to distinguish whether it's due to the AI or the recording conditions.

Common Errors and How to Fix Them

Hands-On 2: Comparing accuracy in English and Thai

Here is a summary of common things that often happen around recording.

Basic authentication appears multiple times

Incorrect ID / password input

- Check the details on the supporting website
- Check for full-width characters

Can't use a microphone

Rejecting in the browser

- Reset site permissions
- See separate steps for iOS / Android

Upload fails

Recording is too long

- Re-record in 1–2 minutes

Transcription results are empty

Silence / Recording Failure

- Play audio to check
- Approach the microphone and re-record

Please look for error messages directly on the supporting website's Troubleshooting section.

Progress Check

Checkpoint

Before moving on, please check if you have made this far.

 **Able to transcribe sample English audio**

 **Could transcribe sample Thai audio**

 **Mistakes were counted against the script**

 **Filled out 4 items on the worksheet**

How to proceed

If you're already finished, go to Optional. If something isn't going well, please ask the instructor or their representative.

Overview (3) (55-63 min)

Transcription accuracy

Transcription Is Not Always Accurate

No matter how good a model is, mistakes will inevitably be made. First, it's important to "check."

proper noun

- Personal Name, University Name, Product Name
- Unknown words can turn into similar sounds

Numbers & Dates

- 15 and 50, Wednesday and Thursday
- Easy to mishear, especially when spoken quickly

word with the same sound

- English 'two / to/too
- Judgment is based on context, but sometimes it can be wrong

hesitation

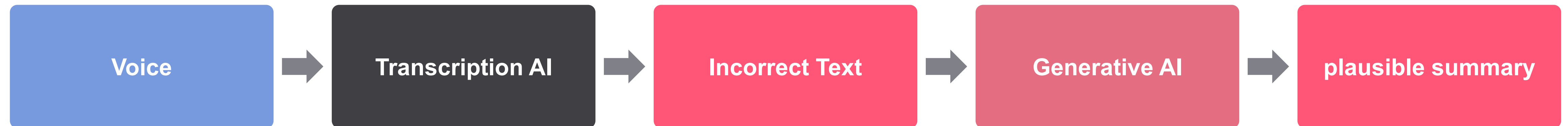
- "Um," "uh" (filler words)
- Turns into text as is

How to check

The most reliable and easiest way to understand is to compare the script (correct answer) sentence by sentence.

Transcription Errors Carry Over to Later Steps

If the transcription is wrong, the summary builds the sentence based on that error.



What will happen?

- Transcription errors cannot be fixed during the summarization stage

And it's hard to notice.

- The writing feels natural

So, before summarizing, check the transcript. This is the most important point today.

Another limitation: lack of speaker information

When recording with multiple people, the speaker's information is not recorded in the text.

Three Things You Need for Minutes

- Decisions — Easily identified
- Deadlines — Easily identified
- Action Item Owners — Often missed

Why can't it be taken?

- The transcription results are a single piece of text with speakers mixed in.

See for Yourself in the Development Exercise

If you ask for a "person responsible" from text without speaker information, what does generative AI do?

In the hands-on (3) development tasks, you can actually see what actually happens.

MISSION 03

Check the transcription and summarize

Required

1. Compare the English transcription results with the script sentence by sentence
2. Correct errors by hand to make the text correct
3. Summarize the revised text using generative AI
4. Compare how the summary changed before and after revision

Optional | Extra Challenge

- Try the same with the audio you recorded yourself
- Output in meeting-minutes format (decisions, owners, deadlines)
- Splitting long audio and transcribing it

You'll use the English transcription result from Hands-On 2.

STEP 1 | Compare the script and sentence by sentence

Hands-On 3: Check the transcription and summarize

The English transcription results obtained in hands-on (2) are compared with the script on a supporting website.

- 1 Copy the transcription results displayed in WSL**
- 2 Paste it into the comparison page of the supporting website**
- 3 Compare the script and sentence by sentence, from top to bottom**

Order of comparison

- First, proper nouns
- Next, numbers and dates
- Finally, the meaning of the entire sentence

How to copy from the WSL terminal

1. Drag and select the area you want to copy with your mouse
 2. Right-click (or Ctrl+Shift+C) to copy
- * Ctrl+C cannot be used as it is used to interrupt processing.
3. Paste into the input field of the supporting website (Ctrl+V)

STEP 2 | Writing Misrecognition on Worksheets

Hands-On 3: Check the transcription and summarize

Correct Text

How was it written in the script?

Transcription Result

How was it perceived?

Types

Proper nouns / Numbers / Homophones / Others

Example Entries (from sample_long_en.wav, for Reference)

Script (correct writing)	Transcription Results	Details of the Differences
The Kruger National Park (KNP)	The Kruger National Park	Abbreviations (if not pronounced aloud, it is normal)
north-east	northeast	Presence or absence of hyphens (inconsistencies)
east, Zimbabwe	east Zimbabwe	missing comma (punctuation)
is the Crocodile River	in the Krakadar River	misidentification (is→in, Crocodile→ Krakadar)
19,500 km²	19,500 kilometers	Unit error (km² → kilometers)
studying information	sending information	Misidentification (mixing up words with similar sounds)

You don't have to write everything down. Refer to the example above and record any mistakes you notice on your worksheet.

STEP 3 | Correct errors to make the text correct

Hands-On 3: Check the transcription and summarize

Errors you write out are manually corrected and saved to sample_long_en.txt.

1 Create a new sample_long_en.txt with nano

2 Paste transcription results and correct errors one by one.

3 Save and check the contents

Save the corrected text to a file

```
cd ~/02_AI_ASR
nano sample_long_en.txt
# Right-click to paste → Correct errors
# Save: Ctrl+O → Enter | Exit: Ctrl+X
cat sample_long_en.txt
# Check if the contents were saved correctly
```

Why fix it?

If you summarize text that still has errors, you get a summary built on those errors instead of the truth.

STEP 4 | Summarize the revised text

Hands-On 3: Check the transcription and summarize

Reading and summarizing sample_long_en.txt

```
python3 -c "  
import os, json, urllib.request  
token = os.environ.get('AI_ENGINE_TOKEN')  
text = open('sample_long_en.txt', encoding='utf-8').read()  
payload = {'model': 'gpt-oss-120b',  
           'messages': [{'role': 'user',  
                           'content': f'Summarize this in English:¥n¥n{text}'}]}}  
req = urllib.request.Request(  
    'https://api.ai.sakura.ad.jp/v1/chat/completions',  
    data=json.dumps(payload).encode('utf-8'),  
    headers={'Authorization': f'Bearer {token}',  
            'Content-Type': 'application/json'})  
with urllib.request.urlopen(req) as res:  
    print(res.read().decode('utf-8'))  
"
```

Key Points

sample_long_en.txt The contents are automatically loaded and sent.

This is the same API used in Module 1 chat/completions.

STEP 5 | Compare before and after the correction

Hands-On 3: Check the transcription and summarize

Using the same steps, I will summarize and compare the text that hasn't been corrected.

Summary of unedited text

- Misidentified words remain as they are
- Turning incorrect content into natural-sounding writing
- The errors aren't obvious just from reading it

Summary of the revised text

- Proper nouns and dates must be entered correctly
- The content matches the script
- Quality that can be shared directly

If you don't fix the previous process, you can't fix the later one. This is the most important discovery today.

Development | Output in Meeting-Minutes Format

Hands-On 3: Check the transcription and summarize

By changing the summary instructions, you can also format the output as meeting minutes.

Stack instructions

Summarize as meeting minutes.

Output in this format: Decisions / Action items / Due dates.

Keep it under 150 words.

Decide the format

- By specifying the output heading, it can be displayed in the same format every time

Decide the length

- Keep the summary at a readable length

Compare with the script

- Whether decisions and deadlines are included correctly

Development | What Happens When Speaker Information Is Missing?

Hands-On 3: Check the transcription and summarize

When multiple people are recorded together, figuring out "who is responsible" becomes tricky.



Possible Outcomes

- Blank / Return 'Unknown' / Produce a guessed result
- How it will turn out depends on the model and instructions

Always Verify

- Do not trust the output as-is; compare it with the script or original text

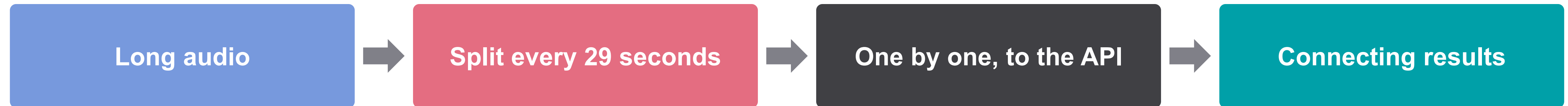
To Prevent Guessing, Say So in the Instructions

If the speaker cannot be identified, leave the person field blank. Do not guess.

Development | Handling audio over 30 minutes long

Hands-On 3: Check the transcription and summarize

Long meetings are split and sent multiple times to the API.



Processing unit inside the model: approximately 30 seconds

- Whisper processes audio for about 30 seconds each
- The script is divided into 29-second segments to match this

API limit: 30 minutes / 30MB

- This is the maximum amount you can send per request.
- If you exceed the limit, you will split it yourself and send multiple times

"30 seconds" and "30 minutes" are different matters

30 seconds is the unit of internal processing for the model, and 30 minutes is the API transmission limit. Please be careful not to confuse them.

Development | Preparing a Python Runtime Environment

Hands-On 3: Check the transcription and summarize

To run scripts, a dedicated Python environment is created.

Running on WSL (Ubuntu)

Note: Some of these may already be set up in your environment — skip any step that's already done.

```
cd ~/02_AI_ASR
sudo apt update
sudo apt install python3-venv ffmpeg
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install requests pydub
```

venv

- Separate from other environments,
- Manage the library

requests / pydub

- API communication,
- Use for splitting audio files

ffmpeg

- pydub used internally
- Voice Conversion Tool

Development | Contents of the Split Script

Hands-On 3: Check the transcription and summarize

splitmp3.py (Excerpt)

```
API_URL = "https://api.ai.sakura.ad.jp/v1/audio/transcriptions"
MODEL   = "whisper-large-v3-turbo"

audio = AudioSegment.from_file(input_path) # Import audio
for i in range(0, len(audio), chunk_ms): # Cut every 29 seconds
    chunk = audio[i:i + chunk_ms]
    chunk.export(out_file, format="mp3")

files = {"file": (path.name, f, "audio/mpeg")}
resp  = requests.post(API_URL, headers=headers,
                      files=files, data={"model": MODEL})
```

1

Split audio every 29 seconds

2

Send each item to the API

3

Merge and save results

Output files

transcript.txt

transcript_full.txt

transcript.json

Development | Putting It Into Action

Hands-On 3: Check the transcription and summarize

Run the script

```
python3 splitmp3.py --input sample_long_en.wav
```

Running display (example)

```
[Split] output_000.mp3 (00:00 - 00:29)
```

```
[Split] output_001.mp3 (00:29 - 00:58)
```

```
[POST] output_000.mp3 (00:00-00:29) [1/4]
```

```
[OK] output_000.mp3 -> 140 chars
```

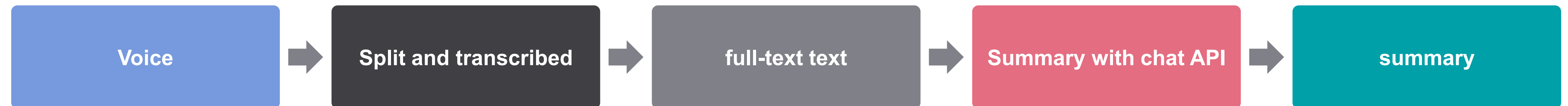
```
[POST] output_001.mp3 (00:29-00:58) [2/4]
```

```
[OK] output_001.mp3 -> 146 chars
```

Development | Combining transcription and summarization

Hands-On 3: Check the transcription and summarize

By adding summarization processing to the script, you can create everything from audio to summaries all at once.



splitmp3+summary.py

```
CHAT_URL = "https://api.ai.sakura.ad.jp/v1/chat/completions"
```

```
CHAT_MODEL = "gpt-oss-120b"
```

```
python3 splitmp3+summary.py --input sample_long_en.wav
```

```
# → transcript_full.txt and transcript_summary.md can be created
```

Tips for Summarizing Long Sentences

First, break it down into short segments and summarize, then summarize that summary again (map-reduce).

Review

Try to see if you can explain the following three points in your own words.

Q1 Where did you tend to make mistakes with transcription?

Q2 How did the summary change before and after correcting the error?

Q3 At which stage do you need to check with people?

If you don't fix the previous process, you can't fix it in the next one. That's why you need a step to check.

Processes that can be automated / Processes that people check

If I organize the system I created today step by step, it looks like this.



Leave it to AI

- Turn audio to text
- Summarize long sentences
- Shape it to a set shape

Person checks

- Whether proper nouns and numbers are correct
- Has the meaning changed?
- Check for missing content

Design is not about "leaving everything to you" or "doing everything," but about deciding the scope of delegation.

API used today

We connected and used SAKURA AI Engine two APIs in order.

/v1/audio/transcriptions

- Convert audio files to text
- Send files using multipart/form-data
- Model: whisper-large-v3-turbo

/v1/chat/completions

- Summarize the text
- Send prompts in JSON
- The same API used in Module 1

By combining APIs, you can create a single flow from voice to summarization.

Practical Use

Today's system can be directly applied to various situations.

Preparation of Minutes

- Record the meeting and share it on the same day

Call center

- Convert call records to text and use them to analyze service quality

Video subtitles

- Add subtitles to lectures and explanatory videos

All of these solve the same problem: "You can't search or analyze using voice alone."

What lies ahead

The system we created today expands even more when combined with other features.

Registering transcriptions on RAG

- If you register for RAG, which you learned at Module 1,
- Enables search for past audio

Use together with images

- Have it read photos of the whiteboard as well.
- This is the multimodal capability covered in the next session, Module 3

By combining the APIs you learn one by one, you can create a system that can be used in actual business operations.

Summary

Thank you for your hard work. This is what you have learned to do in this class.



Ran the speech transcription API using curl



Explain the process and limitations of passing audio to the API



Verified transcription accuracy by comparing it with the script



Explain how errors affect the summary

After this, with a break, we proceed to Module 3. Next is multimodal work, which deals with images.

Account Tokens and environment variables will be used as they are in the classes that follow. Do not delete them.